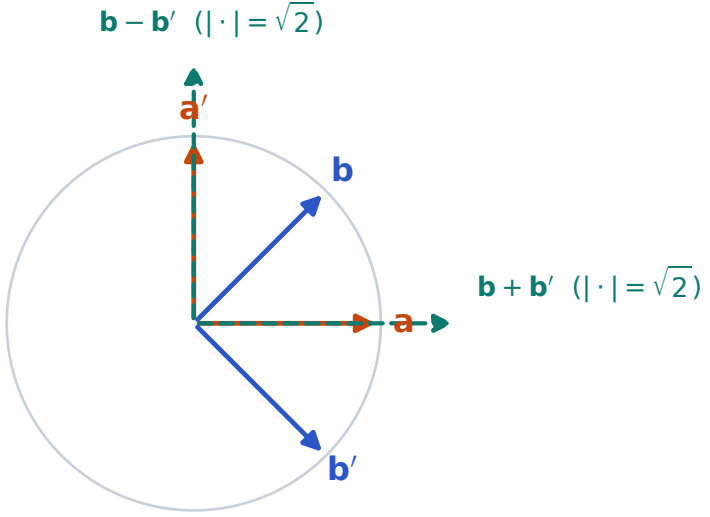
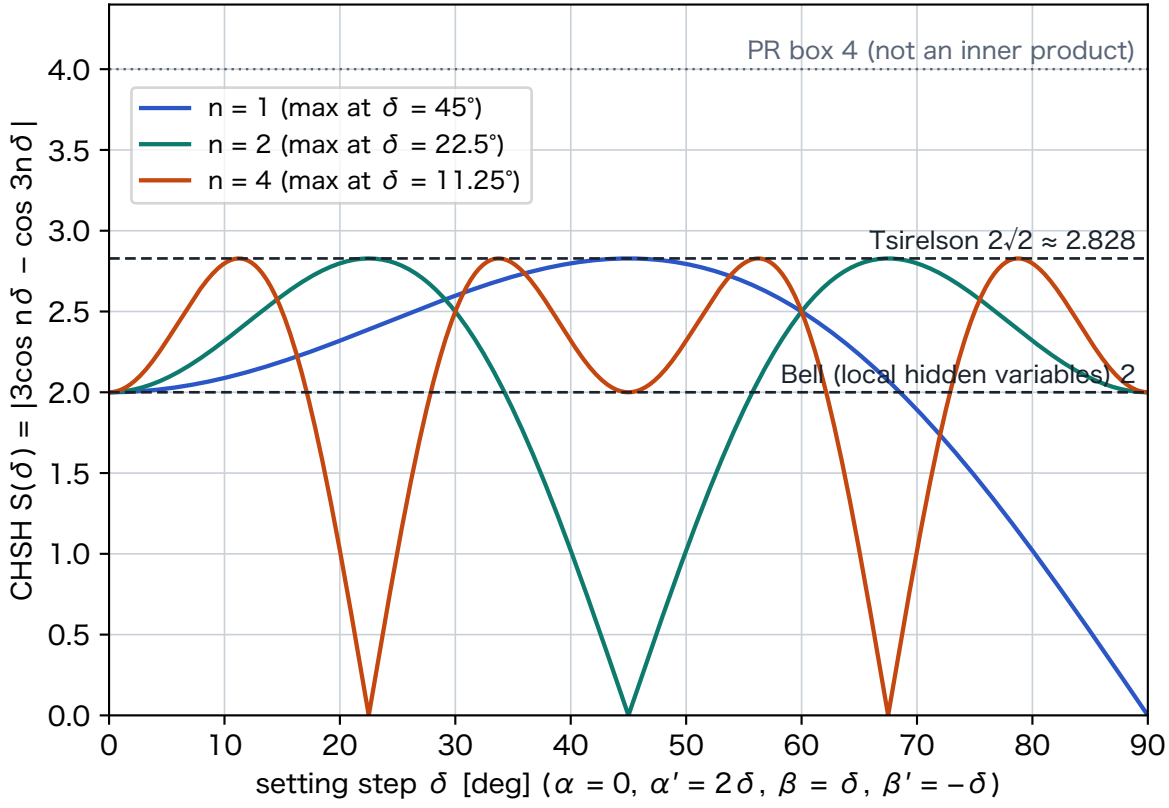
Fig. 4 Tsirelson bound (3.4): if the correlation is an inner product of unit vectors, CHSH never exceeds $2\sqrt{2}$

(a) unit vectors in the readout plane (angles times n)



E = −a·b. S = a·(b+b′) + a′·(b−b′) = √2 + √2 = 2√2
equality: b ⊥ b′, a ∥ b+b′, a′ ∥ b−b′ (nα = 0, nα′ = 90°, nβ = ±45°)

(b) S never exceeds $2\sqrt{2}$; the maximizing step is 45°/n